OPERATING SYSTEM
ORANGE PROBLEM 2

Date : 19/04/26

Semester : 4
Section : I

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Screenshots →

Screenshot 1A :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 1B :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 2A :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ xxd
```

Screenshot 2B :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ xxd ".pes/objects/5d/d26712db190a6306770d643b614a6b23f0df6a1fed469a6f7bec76dc6ffabd" | head -20
00000000: 7472 6565 2034 3900 3130 3036 3434 2064  tree 49.100644 d
00000010: 756d 6d79 2e74 7874 000b d690 98bd 9b9c  ummy.txt........
00000020: c593 4a61 0ab6 5da4 29b5 2536 1147 faa7  ..Ja..].).%6.G..
00000030: b5b9 2291 9e9a 2314 3d                   ..".#.=
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 3A :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  Makefile
  untracked:  README.md
  untracked:  commit.c
  untracked:  commit.h
  untracked:  index.c
  untracked:  index.h
  untracked:  object.c
  untracked:  pes.c
  untracked:  pes.h
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  test_tree.c
  untracked:  tree.c
  untracked:  tree.h

siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 3B :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776543414 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776543414 6 file2.txt
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 4A :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ ./pes log
commit 3b2fc4f6dab2c7a321e072de5f2200318348fe7df67ff785124b424996e25c41
Author: PES User <pes@localhost>
Date:   1776543658

    Add farewell

commit 1c262969fac77d5f6aca51299c0ad6a3e4d773685513d2ecfd957f081bf68d4c
Author: PES User <pes@localhost>
Date:   1776543658

    Add world

commit 75069383700b412fe3ec6aa98de816c890e9b49c1ad5b7a23c12f97abc18b79d
Author: PES User <pes@localhost>
Date:   1776543658

    Initial commit

siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 4B :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/1c/262969fac77d5f6aca51299c0ad6a3e4d773685513d2ecfd957f081bf68d4c
.pes/objects/3b/2fc4f6dab2c7a321e072de5f2200318348fe7df67ff785124b424996e25c41
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/75/069383700b412fe3ec6aa98de816c890e9b49c1ad5b7a23c12f97abc18b79d
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Screenshot 4C :

```
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$ cat .pes/refs/heads/main
3b2fc4f6dab2c7a321e072de5f2200318348fe7df67ff785124b424996e25c41
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Full Integration Test :

```
Date:   1776543876

    Initial commit


--- Second Commit ---
Committed: 0f0727398740... Update file.txt

--- Third Commit ---
Committed: e2b5e01a7c52... Add farewell

--- Full History ---
commit e2b5e01a7c523ffc784df7ae7e64f263cab87aefc16a0386182f7d93722c02ca
Author: PES User <pes@localhost>
Date:   1776543876

    Add farewell

commit 0f0727398740fee4970f2d40fe862336a62b69118edf171ac27a7e0abfbf8821
Author: PES User <pes@localhost>
Date:   1776543876

    Update file.txt

commit 93a25ae8de7d68da2a31894370ee1599a151bf194184d51113795e0b06794a6b
Author: PES User <pes@localhost>
Date:   1776543876

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
e2b5e01a7c523ffc784df7ae7e64f263cab87aefc16a0386182f7d93722c02ca

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/0f/0727398740fee4970f2d40fe862336a62b69118edf171ac27a7e0abfbf8821
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/93/a25ae8de7d68da2a31894370ee1599a151bf194184d51113795e0b06794a6b
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e2/b5e01a7c523ffc784df7ae7e64f263cab87aefc16a0386182f7d93722c02ca

=== All integration tests completed ===
siddhesh@fedora:~/prsnl_strg/code/os/orange2/PES2UG24CS502-pes-vcs$
```

Engineering Analysis

Q5.1: How would you implement `pes checkout <branch>`?

- **Answer:** To implement branching, creating a branch simply requires creating a new file at `.pes/refs/heads/<branch>` and writing the current commit hash into it. To implement `pes checkout <branch>`, three things must happen:

1. Update `.pes/HEAD` to contain `ref: refs/heads/<branch>`.
2. Read the tree of the commit that the new branch points to.
3. Update the working directory and the `.pes/index` file to exactly match the state of that target tree. *Complexity:* The operation is complex because you must safely overwrite files in the working directory and index without accidentally destroying any unsaved/uncommitted user changes.

Q5.2: How would you detect a "dirty working directory" conflict?

- **Answer:** You can detect this by comparing the file's current state against both the index and the object store. First, use the `stat` system call to check if the file's `mtime` or `size` differs from its entry in the `.pes/index` (meaning it has uncommitted changes). Next, check if the file's hash in the *current* branch's commit tree differs from its hash in the *target* branch's commit tree. If the file has uncommitted local changes AND the file differs between the two branches, checkout must abort to prevent silently overwriting the user's uncommitted work.

Q5.3: What is "Detached HEAD" and how to recover commits made in it?

- **Answer:** A "Detached HEAD" occurs when `.pes/HEAD` contains a raw commit hash instead of a branch reference. If you make commits in this state, they will chain together normally, but they will not update any branch file in `.pes/refs/heads/`. If you then switch to another branch, those new commits become "dangling" or "unreachable" because no branch pointer references them. To recover them, a user would need to find the dangling commit's hash (e.g., via a history log or filesystem traversal) and explicitly create a new branch pointing to that hash.

Q6.1: Garbage Collection and Space Reclamation algorithm?

- **Answer:** I would use a "Mark and Sweep" algorithm. *Mark Phase:* Use a Hash Set to track reachable objects. Start at all branch references in `.pes/refs/heads/` and `.pes/HEAD`. Traverse the commit history backwards, marking every commit visited. For each commit, traverse and mark its associated tree, and recursively mark all sub-trees and blobs referenced within. *Sweep Phase:* Iterate through all files in the `.pes/objects/` directory. If an object's hash is not in the "Marked" Hash Set, delete it using the `unlink()` system call. *Estimation:* For 100,000 commits, because identical files and unmodified directories share the same blobs/trees across commits (deduplication), the number of unique objects visited would be a fraction of the total possible objects, but you would still need to compute/visit hundreds of thousands of links depending on repository churn.

Q6.2: Race condition with GC and how to avoid it?

- **Answer:** *The Race Condition:* A user runs `pes add` which writes a new blob to `.pes/objects/`. Concurrently, GC starts its "Mark" phase. Because the user hasn't run `pes commit` yet, the new blob isn't referenced by any commit or branch, so GC doesn't mark it. GC moves to the "Sweep" phase and deletes the newly created blob. The user then runs `pes commit`, which tries to reference the now-deleted blob, resulting in a corrupted repository. *Prevention:* Real Git avoids this using a "grace period" (e.g., objects modified in the last 14 days are completely ignored by GC) or by relying on filesystem locks to block concurrent writes during pruning.